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REPORT ON DESIGN OF CMOS INVERTER FOR DIFFERENT VALUES OF K_R COMPARING THEIR SWITCHING THRESHOLD AND NOISE MARGIN

Assignment for Sessional Test-II
for the course CMOS Design (EC310)

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SUMMARY:

This report investigates the impact of the sizing ratio (K_R) on the static performance metrics of a CMOS inverter, specifically focusing on the switching threshold (V_{sth}), noise margins (NM_L, NM_H) and propagation delays (t_{plh}, t_{phl}). Three inverters were designed and simulated with targeted K_R values of 0.5, 1.0, and 2.0. The $K_R = 1$ baseline was established empirically to center the V_{sth} at $V_{DD}/2$. DC sweep simulations confirmed that K_R directly controls the symmetry of the Voltage Transfer Characteristic (VTC). The symmetrical design ($K_R = 1$) achieved the optimal balance with $NM_L = NM_H = 0.6589$ V and also $t_{plh} = t_{phl}$. Conversely, the asymmetrical designs significantly compromised the overall noise tolerance, the PMOS-dominant $K_R = 2$ design resulted in a critically low NM_H of 0.0623 V and high t_{phl}. The study concludes that setting $K_R = 1$ is essential for maximizing the minimum noise margin and faster transition, thereby ensuring the most fast, robust and reliable operation of digital logic gates.

Software Used: Cadence Virtuoso

INTRODUCTION:

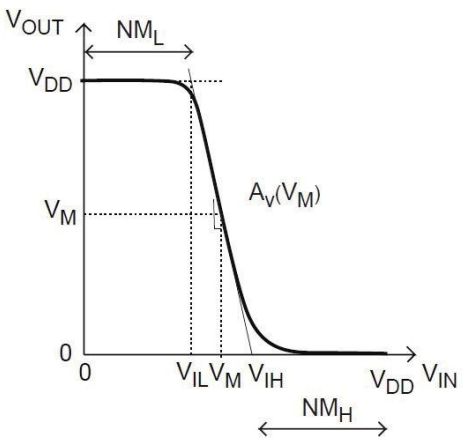
A CMOS inverter in digital integrated circuits is a fundamental building block that performs the NOT logic function using a series combination of a PMOS and an NMOS transistor. It works by inverting the input signal: when the input is low, the PMOS is on and NMOS is off, producing a high output; when the input is high, the NMOS is on and PMOS is off, resulting in a low output.

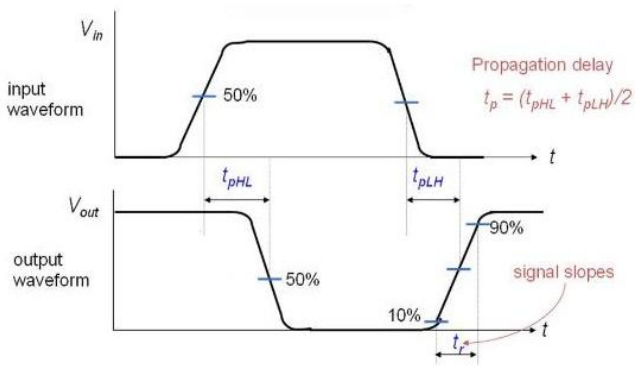
The primary functional metric for any inverter is its **Voltage Transfer Characteristic (VTC)**, which defines the relationship between the input and output voltages.

The performance and reliability of a CMOS inverter are intrinsically linked to the relative sizing the **NMOS pull-down** and the **PMOS pull-up** networks. The process parameters dictates that electrons in the NMOS transistor possess significantly higher mobility μ_n than holes in the PMOS transistor μ_p. This mobility difference means that an NMOS device naturally drives more current than a geometrically identical PMOS device.

This report is about studying the effect of this required sizing adjustment using a variable called the **sizing ratio,** K_r. We will design and test three different inverters with various K_r values (one weak, one balanced, and one strong). Our main goals are to see:

1. How K_r affects the **switching point (**V_{th}**)**—the exact voltage where the inverter flips from High to Low.
2. How K_r changes the **noise margins (**NM_L **and** NM_H**)**—how much noise the inverter can handle.
3. How K_r changes the **propagation delay (**t_{plh} **and** t_{phl}**)**—how much noise the inverter can handle.





BACKGROUND THEORY:

Process Parameters

The current-driving capability of a MOSFET is governed by its process transconductance parameter, k', which is a product of the carrier mobility (μ) and the gate oxide capacitance (C_{ox}).

$K_n = \mu_n C_{ox}$ and $K_p = \mu_p C_{ox}$

The Sizing Ratio (K_R)

To ensure that the inverter's pull-up strength (PMOS) matches the pull-down strength (NMOS), the geometric dimensions of the transistors must be adjusted. The sizing ratio (K_R) is used to quantify the ratio of the effective drive strengths:

$$K_R = \frac{K_p}{K_n} \cdot \frac{(W\ /\ L)_p}{(W\ /\ L)_n}$$

Here, $(W/L)_n$ and $(W/L)_p$ are the aspect ratios (width-to-length) of the NMOS and PMOS transistors, respectively. To achieve symmetry and equal current drive, K_R must be designed to be unity:

$$K_R = 1 \rightarrow \frac{(W/L)_p}{(W/L)_n} = \frac{k_n}{k_p} = \frac{\mu_n}{\mu_p}$$

Switching Threshold (V_{sth})

The switching threshold, V_{sth} is defined as the input voltage V_{in} where the output voltage V_{out} is equal to the input voltage. At this point, the NMOS and PMOS transistors are carrying equal currents $I_{Dn} = I_{Dp}$, and both are operating in the saturation region.

The value of V_{sth} is directly determined by the sizing ratio, K_R

- $K_R = 1$: Symmetrical Voltage Transfer Characteristic (VTC), resulting in $V_{th} \approx \frac{V_{DD}}{2}$.
- $K_R > 1$: V_{th} shifts towards V_{DD}.
- $K_R < 1$: V_{th} shifts towards 0 V.

Noise Margins (NM_L and NM_H)

Noise margins indicates the circuit's tolerance to noise, ensuring reliable operation. They are defined based on four key voltage points extracted from the VTC:

1. V_{OH}: The maximum output voltage (ideally V_{DD}).
2. V_{OL}: The minimum output voltage at 0V.
3. V_{IL}: The maximum input voltage which can be interpreted as logic 0 input.

4. V_{IH}: The minimum input voltage which can be interpreted as logic 1 input.

The noise margins are calculated as:

High Noise Margin: $NM_H = V_{OH} - V_{IH}$

Low Noise Margin: $NM_L = V_{IL} - V_{OL}$

A robust design aims for $NM_H = NM_L$, which is achieved when V_{th} is centered at $\frac{V_{DD}}{2}$. Deviations from $K_R = 1$ will result in unequal noise margins, compromising overall noise immunity.

$$V_{IH} = \frac{V_{DD} + V_{top} + K_R(V_{ton} + 2V_{out})}{1 + K_R}$$

$$V_{IL} = \frac{2V_{out} + V_{top} - V_{DD} + K_R V_{ton}}{1 + K_R}$$

$$V_{sth} = \frac{V_{ton} + \frac{1}{\sqrt{K_R}}(V_{DD} + V_{top})}{1 + \frac{1}{\sqrt{K_R}}}$$

DESIGN AND METHODOLOGY:

The primary goal of the design phase was to create three distinct CMOS inverter configurations. For that we have selected three different values of K_R:

1. $K_R = 1$

2. $K_R = 2$

3. $K_R = 0.5$

Since the precise process transconductance parameters (specifically, the mobility ratio μ_n / μ_p were not explicitly provided by the technology file, the design process for the balanced inverter ($K_R = 1$) was determined empirically through simulation.

3.1 Establishing the Baseline Design ($K_R = 1$)

All designs were implemented using the same gate length, L = 180nm and operated with a supply voltage V_{DD} of 1.8V.

For the estimation of width of pmos for baseline design of $K_R = 1$, we run dc analysis simulations for different values of w_p until we obtain of V_{sth} of approximately 0.9V.

1. **NMOS Sizing:** The NMOS transistor dimensions were held constant across all designs:
    - L_n = 180nm
    - W_n = 850nm
2. **Empirical PMOS Sizing Determination:** The target for the symmetrical design was a switching threshold V_{sth} at the midpoint of the supply voltage: $V_{sth} = V_{DD}/2 = 0.9V$
    - A DC sweep simulation was performed on the testbench, iteratively adjusting the PMOS width (W_p) until the output voltage crossed the input voltage approximately at 0.9V
    - The resulting dimensions for the K_R=1 design were:

      W_p (K_R = 1) = 3.4µm
3. **Determination of Effective Mobility Ratio μ_n / μ_p** : The K_R = 1 condition requires the PMOS to compensate for the lower hole mobility μ_p. The ratio of the NMOS and PMOS widths required to achieve $V_{sth} = V_{DD}/2$ is equal to the ratio of their mobilities:

   Using the empirically determined widths, the **effective process mobility ratio** was calculated and established as the design constant:

$$\frac{\mu_n}{\mu_p} = \frac{W_{p(K_R=1)}}{W_n}$$

$$\frac{\mu_n}{\mu_p} = \frac{3.4 \text{ x } 10^{-6}}{850 \text{ x } 10^{-9}} = 4$$

3.2 Calculating width of pmos for K_R=0.5 and K_R=2

With the process mobility ratio established, the PMOS widths for the remaining inverters were calculated using the K_R equation, ensuring all lengths L remained constant at 180nm:

Value of K_R	Cell Name	Properties
0.5	NMOS	Width, W_n = 850nm Length, L = 180nm
	PMOS	Width, W_p = 1.7µm Length, L = 180nm
1	NMOS	Width, W_n = 850nm Length, L = 180nm
	PMOS	Width, W_p = 3.4µm Length, L = 180nm

2	NMOS	Width, W_n = 850nm Length, L = 180nm
	PMOS	Width, W_p = 6.8µm Length, L = 180nm

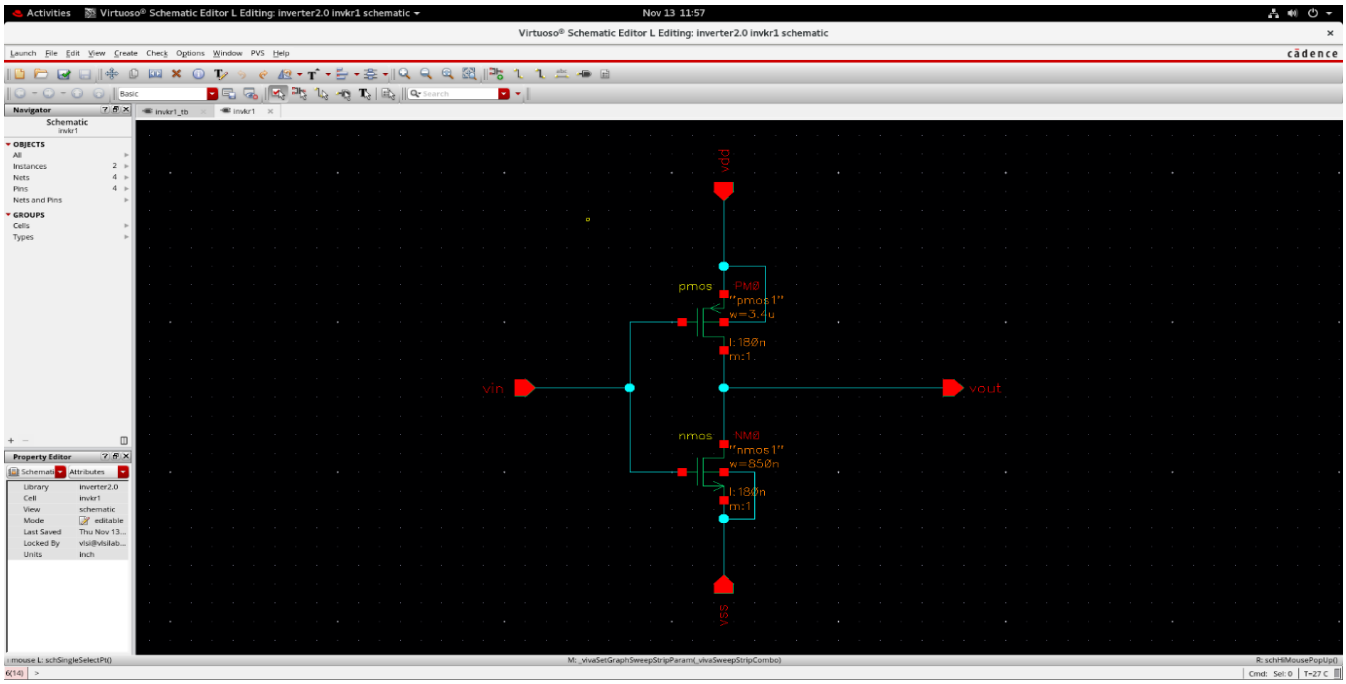


Fig1. Schematic of CMOS inverter for K_R = 1

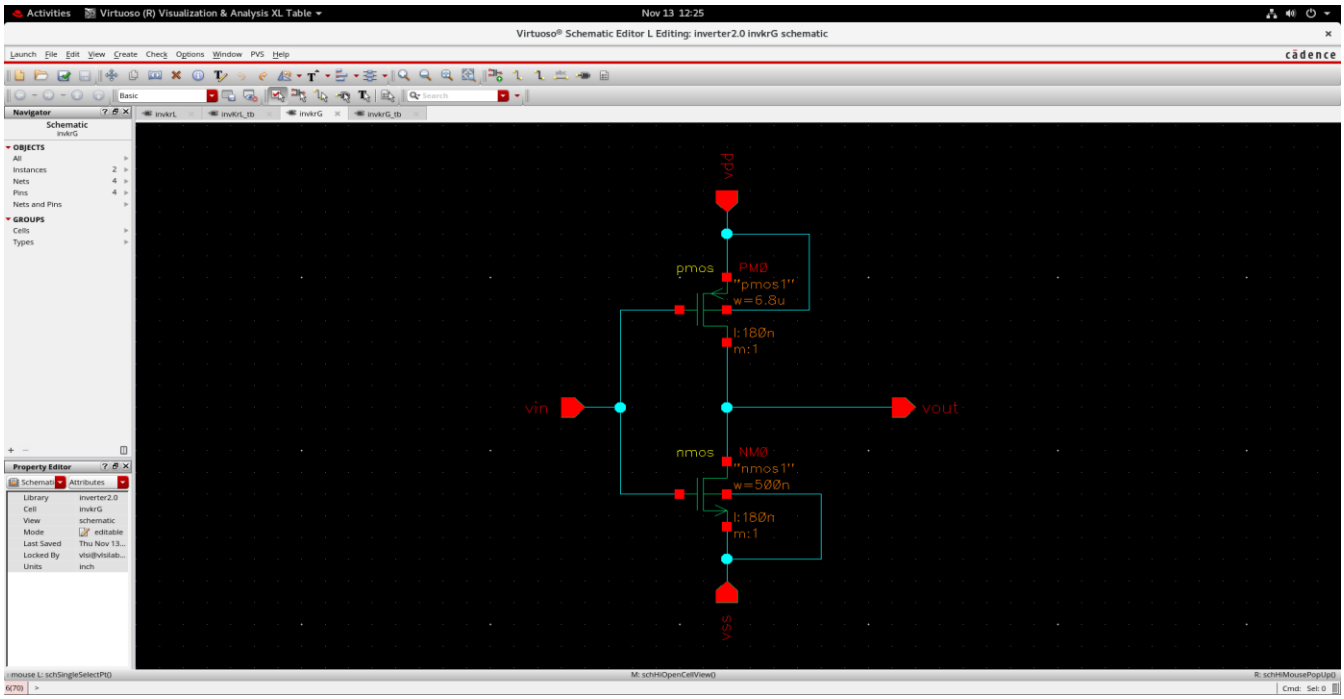


Fig2. Schematic of CMOS inverter for K_R = 2

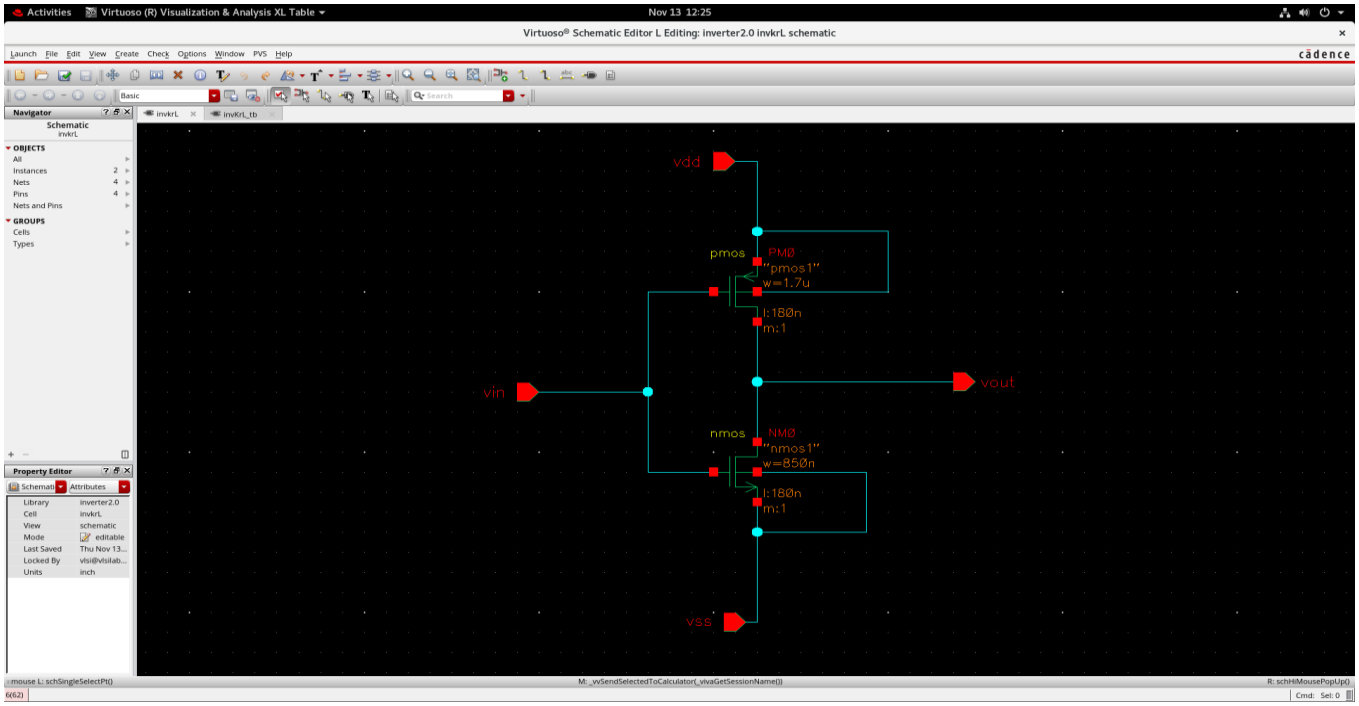


Fig3. Schematic of CMOS inverter for K_R = 0.5

3.3 Simulation Environment and VTC Analysis

All three inverters were designed as an schematic and represented by a corresponding symbol. A testbench schematic was created for each of these symbols to apply the required input and obtain DC analysis and Transient analysis

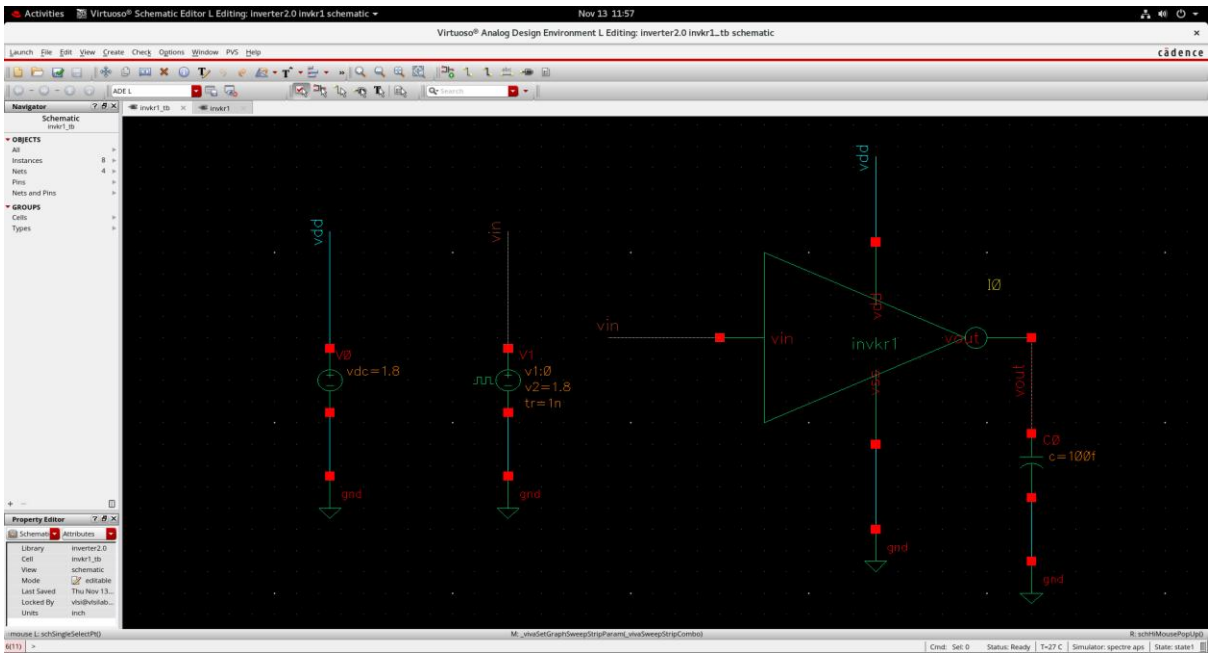


Fig. 4: Testbench for DC and transient analysis of the CMOS inverter

DC Analysis

A DC Analysis was performed for each inverter by sweeping the input voltage from 0V to 1.8V. This produced the Voltage Transfer Characteristic (VTC) plots, from which we can obtain important parameters such as V_{sth}, V_{IL}, V_{OL}, V_{IH}, V_{OH}.

- **Extraction of V_{IL} and V_{IH}:** These critical points were found by taking the derivative of the VTC using the calculator function *deriv* in the ADE L window of Cadence. Then plotting the output and obtaining values intersecting at -1 line.
- **V_{OH} and V_{OL}:** The maximum and minimum output voltages were taken directly from the VTC, using the calculator function *value* at V_{IL} and V_{IH} points in the VTC.

Transient Analysis

A Transient Simulation was also performed by applying a 0V to 1.8V square wave input with a period of 20ns to visually confirm the correct switching action of the designed circuits.
Applied Vin for the transient analysis:
(0V to 1.8V)
Period = 20ns
Delay time = 10ns
Rise time = 1ns
Fall time = 1ns
Pulse width = 10ns

3.4 Noise Margin Calculation

The extracted voltage levels V_{IL}, V_{OL}, V_{IH}, V_{OH} were used to calculate the High Noise Margin NM_H and Low Noise Margin NM_L for each of the three inverter designs:

$NM_H = V_{OH} – V_{IH}$

$NM_L = V_{IL} - V_{OL}$

3.5 Propagation Delay (tphl and tplh) Calculation

A Transient Simulation was performed by applying a 0V to 1.8V square wave input with a period of 1 and a defined capacitive load 100fF. The propagation delay was calculated directly from the resulting waveforms:

- Delay Measurement: The High-to-Low delay tphl and Low-to-High delay tplh were calculated using the built-in delay measurement function within the simulation software's calculator tool.
- tp (Average Propagation Delay): Calculated as the average of tplh and tphl

RESULTS AND DISCUSSIONS:

The graphs obtained after DC and transient analysis are as follows:

(i) For $K_R = 1$:

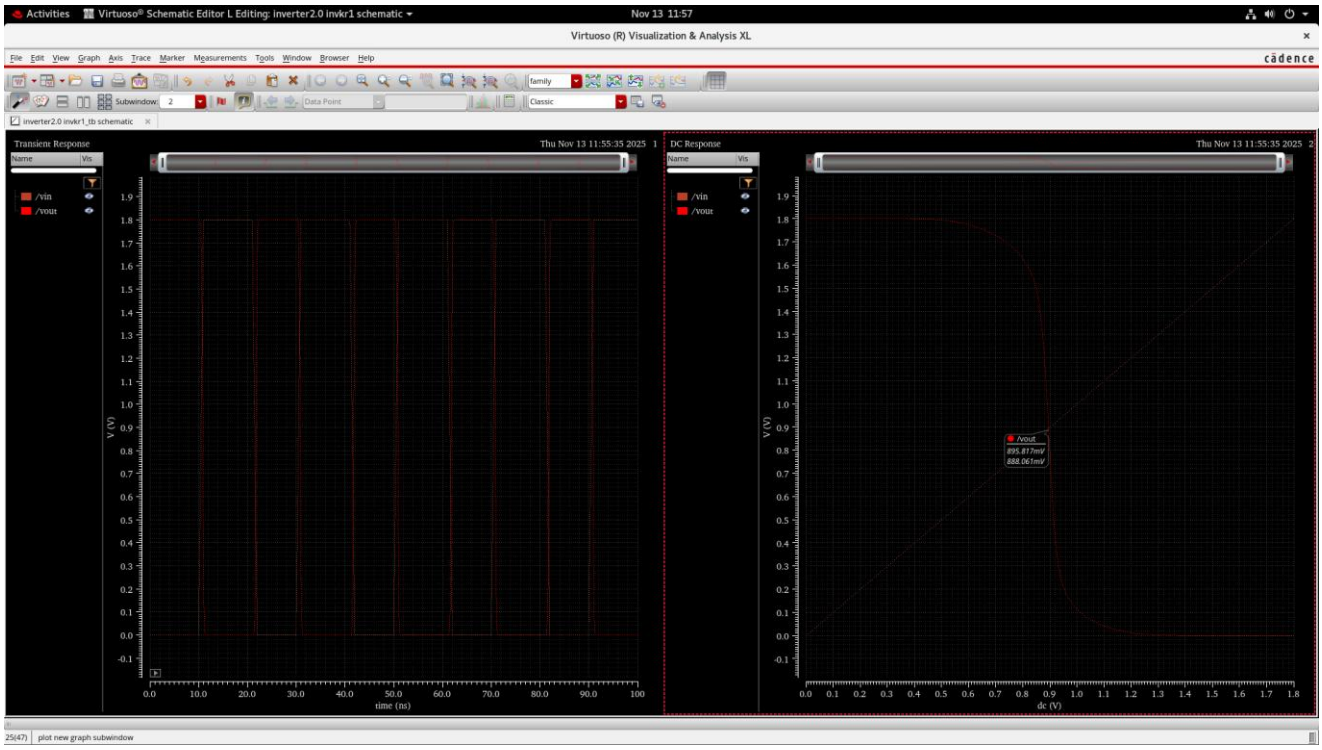


Fig. 5: DC and transient analysis plots for $K_R = 1$ displaying V_{sth} in VTC

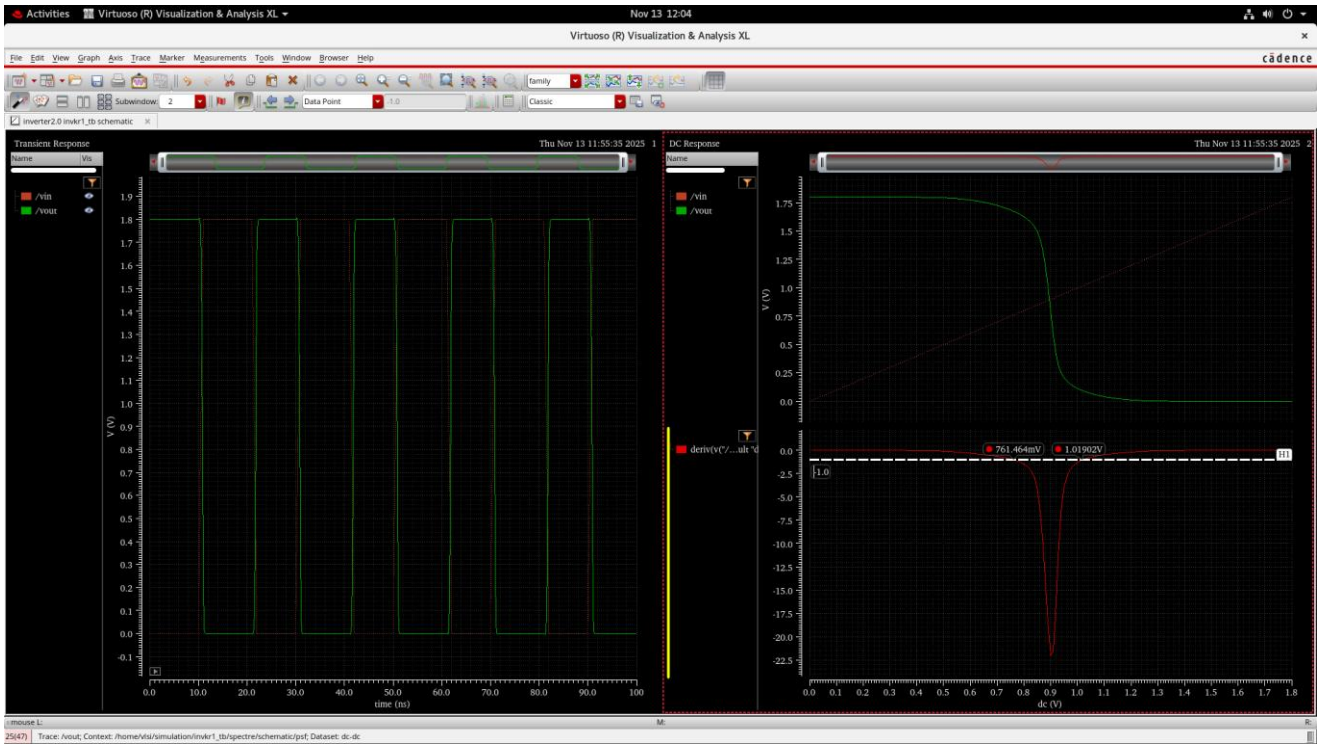


Fig. 6: Plot for slope of VTC showing values of V_{IH} and V_{IL} values for $K_R = 1$

(ii) For $K_R = 0.5$:

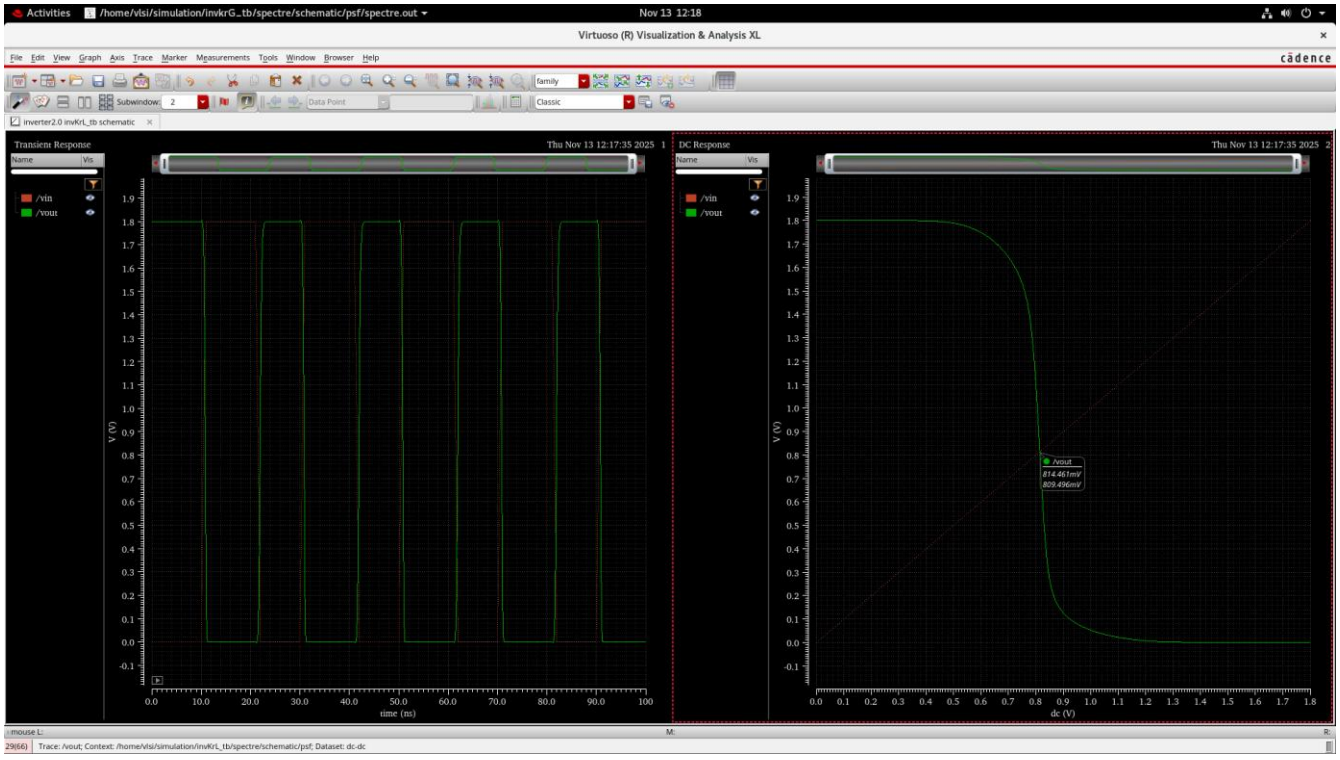


Fig. 7: DC and transient analysis plots for $K_R = 0.5$ displaying V_{sth} in VTC

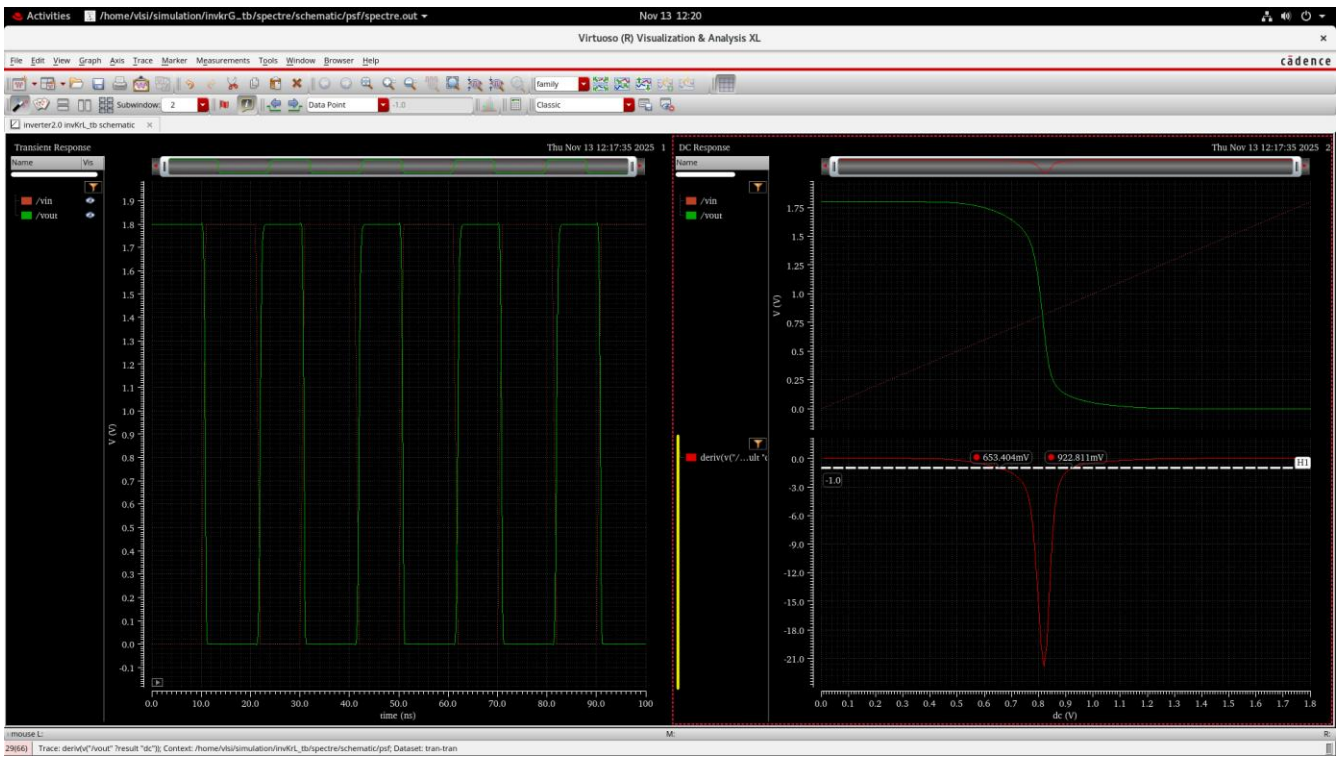


Fig. 8: Plot for slope of VTC showing values of V_{IH} and V_{IL} values for $K_R = 0.5$

(iii) For $K_R = 2$:

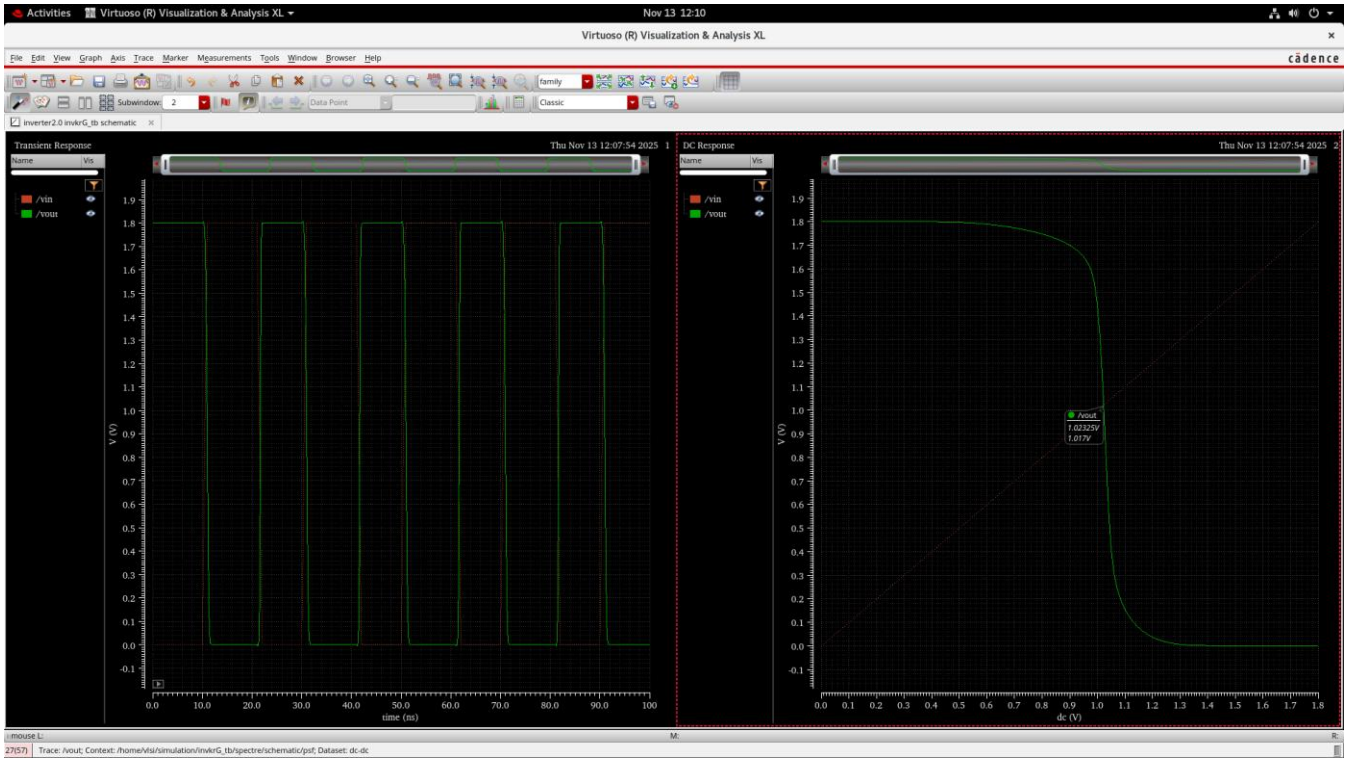


Fig. 9: DC and transient analysis plots for $K_R = 2$ displaying V_{sth} in VTC

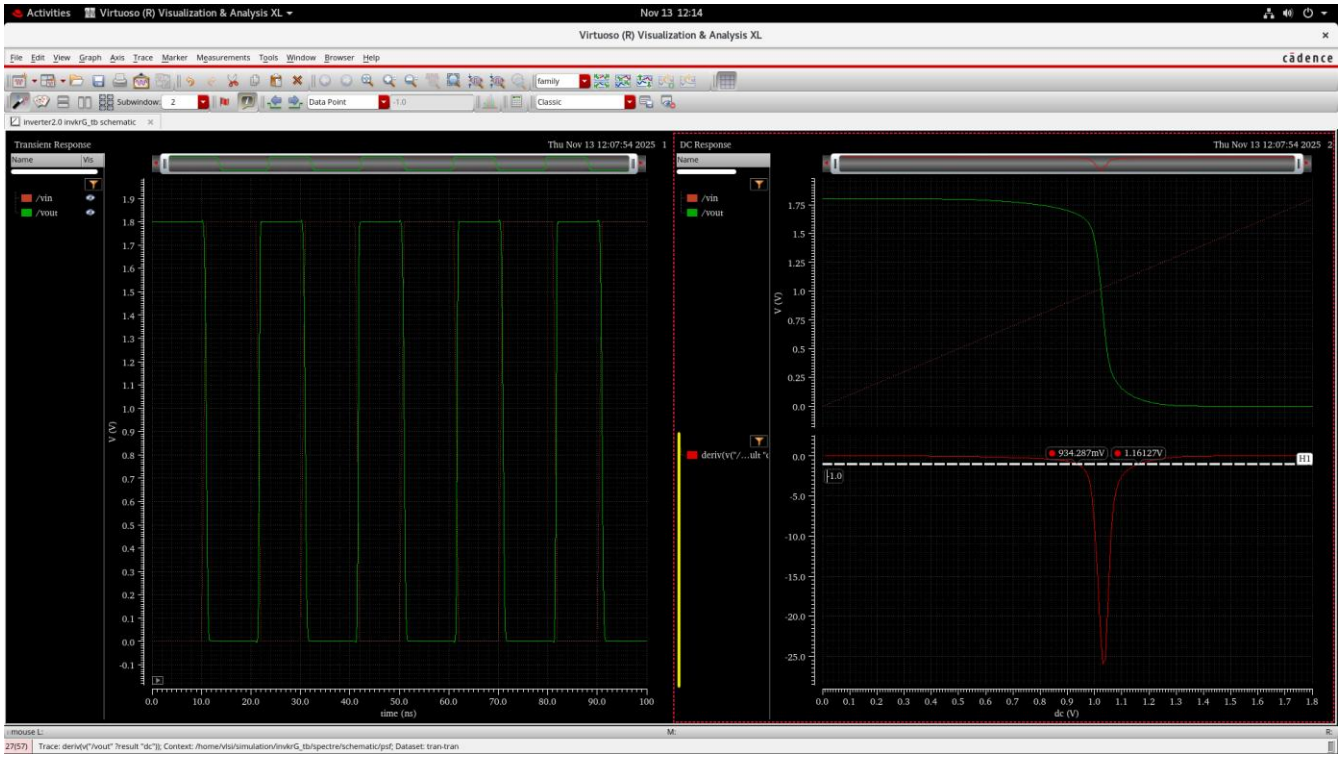


Fig. 10: Plot for slope of VTC showing values of V_{IH} and V_{IL} values for $K_R = 2$

The result obtained from calculator in ADE L for values of V_{OH} and V_{OL}:

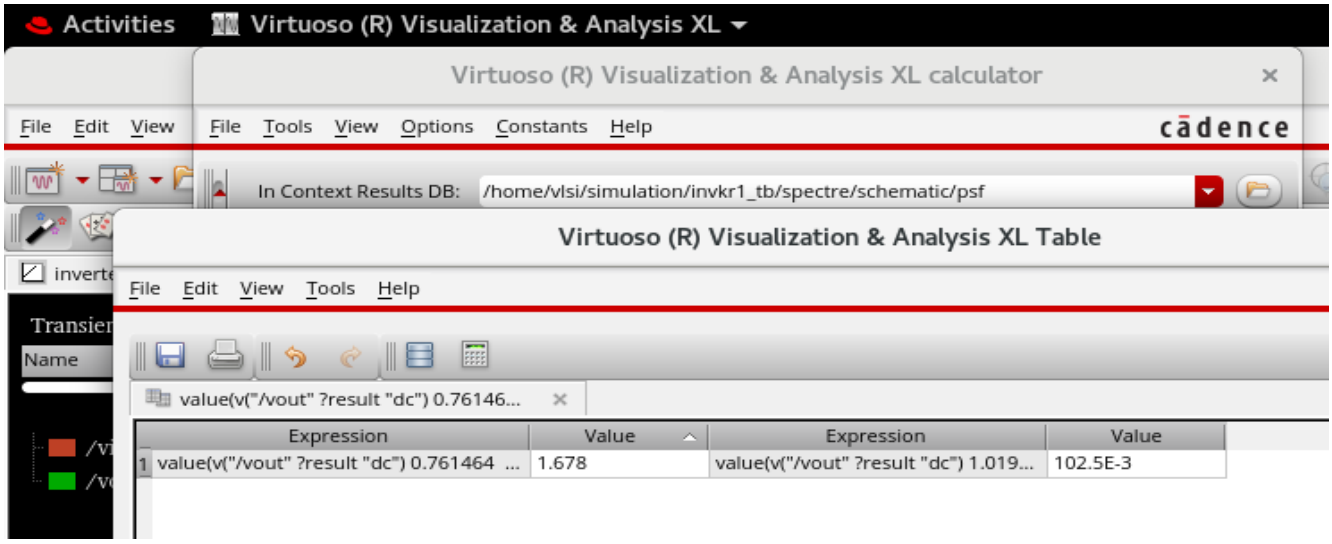


Fig. 11: V_{OH} and V_{OL} output for K_R = 1 using Virtuoso Visualization & Analysis Calculator

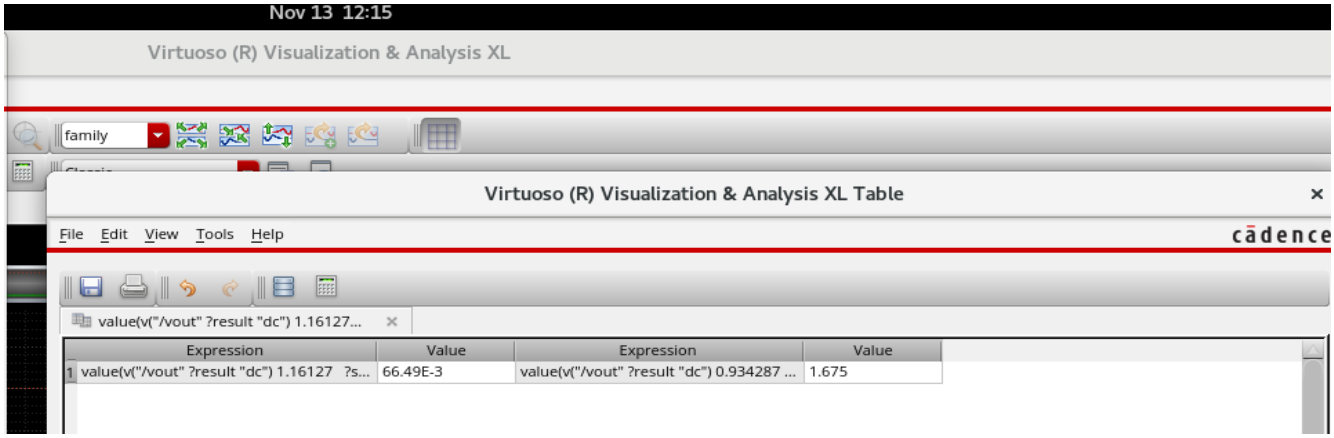


Fig. 11: V_{OH} and V_{OL} output for K_R = 2 using Virtuoso Visualization & Analysis Calculator

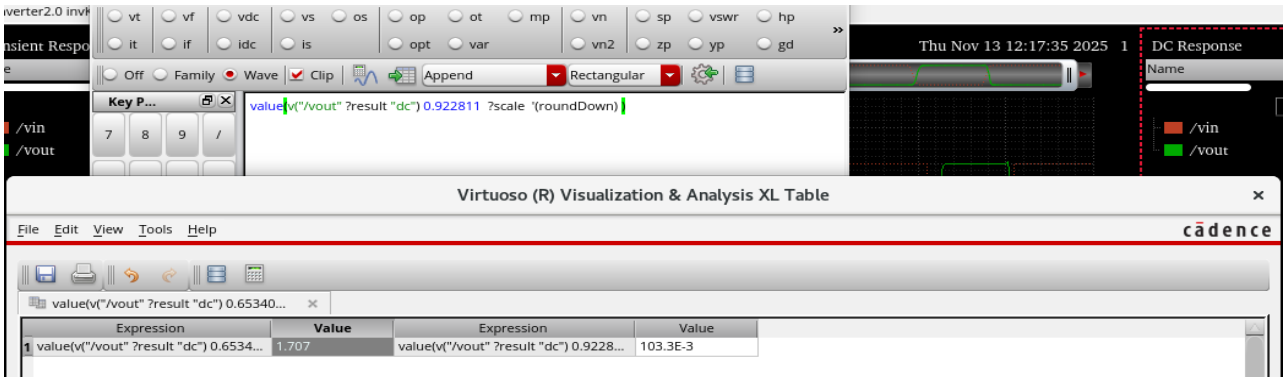


Fig. 12: V_{OH} and V_{OL} output for K_R = 0.5 using Virtuoso Visualization & Analysis Calculator

For the results of **propagation delay** in transient analysis:

Outputs

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	vout		✔		allv
2	vin		✔		allv
3	tplh	469.377p	✔		
4	tphl	335.424p	✔		

Fig. 13: tphl and tplh outputs in transient analysis for K_R = 0.5

Outputs

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	vout		✔		allv
2	vin		✔		allv
3	tphl	351.725p	✔		
4	tplh	317.594p	✔		

Fig. 14: tphl and tplh outputs in transient analysis for K_R = 1

Outputs

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	vout		✔		allv
2	vin		✔		allv
3	tphl	503.556p	✔		
4	tplh	198.373p	✔		

Fig. 13: tphl and tplh outputs in transient analysis for K_R = 2

TABLE OF RESULTS AND OUTPUTS:

Parameters	K_R = 0.5	K_R = 1	K_R = 2
V_{sth}	814.461 mV	895.817 mV	1.0232 V
V_{IL}	653.404 mV	761.464 mV	934.287 mV
V_{IH}	922.811 mV	1.01902 V	1.6127 V
V_{OL}	103.3 mV	102.5 mV	66.49 mV
V_{OH}	1.707 V	1.678 V	1.675 V
NM_L	550.104 mV	658.964 mV	867.797 mV
NM_H	784.189 mV	658.98 mV	62.3 mV
t_{plh}	469.377 ps	351.725 ps	503.556 ps

t_{phl}	335.424 ps	317.594 ps	198.373 ps
t_p	402.4 ps	334.659 ps	350.964 ps

Some important points to derive from the outputs of the simulations:

- For $K_R = 1$, the switching threshold is obtained at 0.8958V which is very close to 0.9V i.e. half of $V_{DD} = 1.8V$.
- For $K_R = 1$, we found out that $NM_L \approx NM_H$ i.e. approximately 0.6589V
- For $K_R = 0.5$, the switching threshold is obtained at 0.814V which is significantly less than 0.9V which is due to stronger NMOS pull causing VTC to shift towards VSS 0V.
- For $K_R = 0.5$, we found $NM_L < NM_H$ which shows that inverter is more vulnerable to noise when we give LOW input.
- For $K_R = 2$, the switching threshold is obtained at 1.0232V which is significantly higher than 0.9V which is due to stronger PMOS pull causing VTC to shift towards VDD 1.8V.
- For $K_R = 2$, we found $NM_L > NM_H$ which shows that inverter is more vulnerable to noise when we give HIGH input and the minimal margin of 62.3mV only suggests extremely poor reliability in practical systems.
- While it is possible to increase one noise margin that is either NM_L or NM_H by moving K_R away from 1, this always comes at the expense of another noise margin.
- We have obtained the lowest transition delay in $K_R = 1$, i.e. the fastest transistor
- The $K_R = 1$ design achieved balanced dynamic performance, resulting in tplh ≈ tphl. This is crucial for avoiding skew in clock and signal timing.
- For $K_R = 0.5$, the stronger NMOS lead to a faster tphl (pull-down) but a significantly slower tplh (pull-up)
- For $K_R = 2$**,** the stronger PMOS lead to a faster tplh (pull-up) but a significantly slower tphl (pull-down)

Based on the above observations, it is showed that design of CMOS inverter with $K_R = 1$ is the optimal choice where maximum noise immunity is required as it provides symmetric noise margin and switching threshold at $V_{DD}/2$.

CONCLUSION:

This report successfully achieved its objective of analysing the effect of the sizing ratio (K_R) on the static performance metrics of a CMOS inverter. By designing and simulating three distinct inverters with K_R values of 0.5, 1.0, and 2.0, the fundamental relationship between transistor sizing, switching threshold, and noise immunity was quantitatively confirmed. The results unequivocally demonstrated that the value of K_R dictates the symmetry of the Voltage Transfer Characteristic (VTC) and, consequently, the balance of the noise margins and transition delay. In conclusion, for designing reliable and fast digital logic inverters that must operate robustly under potential noise interference, the goal must always be to achieve $K_R = 1$ by matching the PMOS width to the NMOS strength according to the process mobility ratio.